

ID U2210251

INHA UNIVERSITY IN TASHKENT (FALL 2023)

Computer Algorithms (CIE3090)

ASSIGNMENT - #3

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Submission Deadline: 09 /October/ 2024 (23:00 Hr)

- > Solve all questions on printout of this page. You are required to solve (**handwritten**) all the questions and submit it on eclass by taking picture of your assignment.
- > Late submissions and email-submissions are not accepted.
- > Submissions **not** conforming to the given format will **NOT** be evaluated.

Q1. Convert the following sequence to a Heap using Build Max Heap
12, 9, 7, 2, 15, 32

Use this space for showing iterations		0	1	2	3	4	5	$i \Rightarrow 2 \text{ to } 0$	
		12	9	7	2	15	32		
$i=2$	Heapify(A, 2)							$L=5$	
	Swap(A[2], A[5])	12	8	32	2	15	7		
	Heapify(A, 5)							Exit	
$i=1$	Heapify(A, 1)							$L=3$	$R=4$
	Swap(A[1], A[4])	12	15	32	2	8	7		
	Heapify(A, 4)							Exit	
$i=0$	Heapify(A, 0)							$L=1$	$R=2$
	Swap(A[0], A[2])	32	15	12	2	8	7		
	Heapify(A, 2)							$L=5$	
								Exit	

Answer: 32, 15, 12, 2, 8, 7

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Q2. Apply Quicksort Algorithm to sort the given sequence

(Show only steps of first PARTITION on the initial array)

Use this space for any calculations	1	2	3	4	5	6
$p=10 \quad i=-1$	9	7	2	15	32	10
$i=0 \quad \bar{j}=0$	8	7	2	15	32	10
$i=1 \quad \bar{j}=1$	8	7	2	15	32	10
$i=2 \quad \bar{j}=2$	8	7	2	15	32	10
$i=2 \quad \bar{j}=3$	8	7	2	15	32	10
$i=2 \quad \bar{j}=4$	8	7	2	15	32	10
	8	7	2	10	32	15

Q2. Apply Merge sort Algorithm to sort the given sequence

(Show all steps)

[illegible]